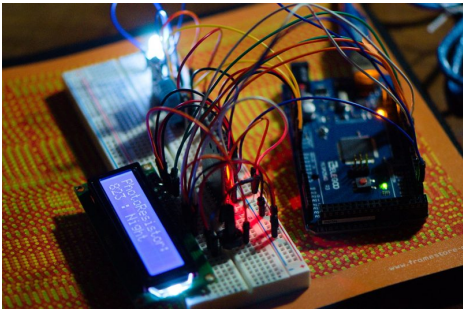


Electronics Tinker Kit

WHAT'S IN THE KIT

- 3 Arduino Uno kits
- Electronics Fun Kit
- Breadboards



QUICK-START GUIDE

1. Start with the breadboard: power rails outside, components inside.
2. Build a simple LED + resistor circuit before adding the Arduino.
3. Use the Arduino IDE (or Tinkercad Circuits online to prototype first).
4. Upload the classic Blink sketch, then modify it.

FULL LESSONS IN THIS GUIDE

GRADES 6-8	Tinkercad-to-Table	page 2
GRADES 9-12	Sensor Alarm Build	page 3

PRO TIP: Tinkercad Circuits (free, browser) lets the whole class simulate before touching hardware.

Electronics Tinker Kit

LESSON · GRADES 6-8

Tinkercad-to-Table

LEARNING OUTCOMES

- Students will build a working circuit on a breadboard from a diagram
- Students will prototype a circuit digitally before physical assembly
- Students will identify and fix common wiring errors

MATERIALS

- Electronics Tinker Kit (breadboards, LEDs, resistors, jumpers)
- Chromebooks with tinkercad.com/circuits (free)
- Printed circuit diagrams
- Troubleshooting checklist

INSTRUCTIONS

1. Teach breadboard anatomy: rails, rows, and what connects to what.
2. Students build the LED circuit in Tinkercad Circuits first and simulate it.
3. Transfer the working digital design to the real breadboard.
4. When circuits fail (they will), use the checklist: polarity, row alignment, resistor present.
5. Add a switch or second LED once the base circuit works.
6. Exit: photograph the working circuit and label current path.

ACTIVITY

Tinkercad-to-Table — simulate first, then build. Comparing the digital circuit that 'worked' with the physical one that doesn't is the fastest misconception-finder in electronics.

Electronics Tinker Kit

LESSON · GRADES 9-12

Sensor Alarm Build

LEARNING OUTCOMES

- Students will design a sensor-triggered alarm system from requirements
- Students will integrate an Arduino with breadboard components
- Students will evaluate a prototype and propose a v2 with justification

MATERIALS

- Arduino Uno kits + breadboards + component assortment
- Arduino IDE or Tinkercad Circuits
- Requirements card (trigger type, response, constraint)
- Design review rubric

INSTRUCTIONS

1. Deal requirement cards: light-triggered, motion/tilt-triggered, or tripwire alarm.
2. Teams sketch a system diagram and get a 2-minute design review before parts.
3. Build the sensing circuit; test sensor readings via serial monitor.
4. Code the trigger logic with a sensible threshold and an alarm response (buzzer/LED).
5. Adversarial testing: another team tries to defeat the alarm.
6. Write the v2 memo: what the attack revealed and what they'd change.

ACTIVITY

Sensor Alarm Build — engineer an alarm to spec, then watch another team try to beat it. The v2 memo turns the exploit into an engineering requirement.